

Advanced Machine Learning

DATA 642

Lecture 5 — Kernel Ridge Regression

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly

Kernel Ridge Regression

We defined the regularized empirical risk minimization problem as:

$$\min_{f \in \mathbb{H}} J(f) = \sum_{n=1}^N \mathcal{L}(y_n, f(\mathbf{x}_n)) + \lambda \Omega(\|f\|^2) \quad (1)$$

Here, $\Omega(\|f\|^2)$ represents the regularization term, which typically takes the form $\|f\|^2$ in squared norm, where $\|\cdot\|$ is the norm in $\mathbb{H}$.

The notation

In the matrix notation derivation, we defined:

- $\mathbf{y}$ as the vector of targets,
- $\mathbf{X}$ as the design matrix containing the input data, and
- $\mathbf{K}$ as the kernel matrix.

The regularization term $\Omega(\|f\|^2)$ is represented by $\lambda\theta^T\mathbf{K}\theta$ in the matrix notation. Here, θ is the vector of coefficients in the RKHS representation of f .

Setting up the problem and defining the gradient

With these definitions, the regularized empirical risk minimization problem becomes:

$$\min_{\theta} J(\theta) = (\mathbf{y} - \mathbf{K}\theta)^T (\mathbf{y} - \mathbf{K}\theta) + \lambda \theta^T \mathbf{K}\theta \quad (2)$$

To find the optimal θ , we take the derivative of $J(\theta)$ with respect to θ and set it to zero:

$$\frac{\partial J(\theta)}{\partial \theta} = -2\mathbf{K}^T (\mathbf{y} - \mathbf{K}\theta) + 2\lambda \mathbf{K}\theta = 0 \quad (3)$$

Solving for θ

Expanding and rearranging, we get:

$$\mathbf{K}^T \mathbf{K} \theta - \mathbf{K}^T \mathbf{y} + \lambda \mathbf{K} \theta = 0 \quad (4)$$

Grouping terms:

$$(\mathbf{K}^T \mathbf{K} + \lambda \mathbf{I}) \theta = \mathbf{K}^T \mathbf{y} \quad (5)$$

Solving for θ :

$$\theta = (\mathbf{K}^T \mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{K}^T \mathbf{y} \quad (6)$$

or

$$\theta = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{y} \quad (7)$$

Make predictions

Once we have the optimal θ , we can compute the predictions using the kernel matrix $\mathbf{K}$:

$$\hat{\mathbf{y}} = \mathbf{K}\theta \tag{8}$$

This matrix-based approach is computationally efficient, especially when dealing with large datasets, as it allows us to perform computations using matrix operations. Additionally, the regularization term helps to control the complexity of the model and prevent overfitting.